tube. At this, I grew rather despondent and in order to cheer me up Archer suggested that we try to develop the gas chromatogram. He and Synge had outlined the basic tenets of the techniques in their seminal paper on the liquid-liquid chromatogram in 1941. ('A new form of Chromatogram employing two liquid phases' (1941) A. J. P. Martin and R. L. M. Synge, Biochem J. 35, 1358) (see Fig. 1).

This clear and unambiguous suggestion, fortunately for me, had fallen on stony ground, even including those workers active in the area of gas-solid chromatography (N.C. Turner, S. Claesson, N.M. Turkeltaub and C.S.G. Phillips). Their work had been concentrated on the use of adsorbents where the non-linear adsorption isotherms always produced tailing peaks and hence impaired separation when operated as elution chromatograms.

Our first attempts were made using the 4 ft by 4 mm I.D. columns from the abortive crystallisation experiments packed with Kieselguhr but this time carrying 25% by weight of a silicone or paraffin oil as stationary phase. The solutes used were the short chain saturated fatty acids since George Popjak in an adjoining laboratory was working on their biosynthesis. The apparatus was of the simplest, the column at laboratory temperature, a source of nitrogen at constant pressure, a micro pipette of our own making, a test tube containing aqueous indicator and an Erlenmeyer flask of dilute alkali used to force drops into the titration cell. After loading the column with a mixture of acetic and propionic acids, the gas supply was connected and the effluent bubbled into the filtration cell, five drops of alkali were added and the time taken to neutralise them noted on a stop-watch, followed by the addition of five more drops, etc. By plotting the time to neutrality against total volume added, one obtained the integral

151. A NEW FORM OF CHROMATOGRAM EMPLOYING TWO LIQUID PHASES

1. A THEORY OF CHROMATOGRAPHY

2. APPLICATION TO THE MICRO-DETERMINATION OF THE HIGHER MONOAMINO-ACIDS IN PROTEINS

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INTRODUCTION

In most forms of counter-current extraction column the very small drop required for the rapid attainment of equilibrium, and hence for high efficiencies, cannot be used owing to the difficulty of preventing it moving in the wrong direction. In the case of a solid, however, for any reasonable size of particle a filter will prevent movement in any undesired direction. Consideration of such facts led us to try absorbing water in silica gel etc., and then using the water-saturated solid as one phase of a chromatogram, the other being some fluid immiscible with water, the silica acting merely as mechanical support. Separations in a chromatogram of this type thus depend upon differences in the partition between two liquid phases of the substances to be separated, and not, as in all previously described chromatograms, on differences in adsorption between liquid and solid phases.

The difficulties of using chromatograms are very greatly lessened when the substances to be separated are coloured, or if colourless can be made visible. Various methods have been used for this [cf. Zechmeister & Cholnoky, 1936; Cook, 1941], though none of these was suitable for our problems. As the substances which we desired to separate were acids, and water was one of our phases, we were able to obtain visual evidence of the presence of any of these acids by adding a suitable indicator to the water with which the gel was saturated.

In the present paper we present an approximate theory of chromatographic separations, and describe an application of the new chromatogram to the micro-determination of the higher monoamino-acids in protein hydrolysates. This method is based on the partition of acetamino-acids between chloroform and water phases, and supersedes the macro-method described by us [Martin & Synge, 1941, 1], being rapid and economical both of materials and of apparatus.

The mobile phase need not be a liquid but may be a vapour. We show below that the efficiency of contact between the phases (theoretical plates per unit length of column) is far greater in the chromatogram than in ordinary distillation or extraction columns. Very refined separations of volatile substances should therefore be possible in a column in which permanent gas is made to flow over gel impregnated with a non-volatile solvent in which the substances to be separated approximately obey Raoult's law. When differences of volatility are too small to permit of ready separation by these means, advantage may be taken in some cases of deviation from Raoult's law, as in azeotropic distillation.

Fig. 1. The first paper in which gas-liquid chromatogram was suggested.

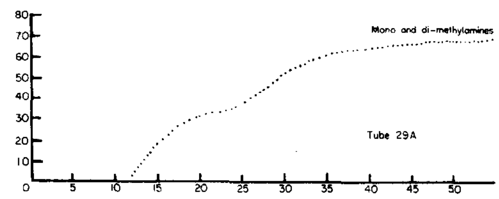

Fig. 2. The first successful gas-liquid chromatogram, the separation of mono- and di-methylamine, detection by manual titration.
0,10 80 Drops of standard alkali.
5,10 50 Time in minutes.

of the normal chromatographic peak.

However, to our surprise the chromatographic zones were distorted, having an extended front and a sharp back, each zone beginning where the previous one ended. We were baffled at first and thought some odd adsorption effect was occurring on the Kieselguhr. We then tried every solid support we could think of (from sand to powdered graphite) without success. We even constructed a 2 mm capillary column using an extended spring to stir the liquid phase on the wall, but still we got distorted zones. Clearly there was some phenomenon occurring that we had not recognized and we were despondent. Almost as a last resort, I suggested we try substances other than fatty acids and so without much hope we attempted a separation of mono- and di-methylamines. To our surprise and joy we got an almost perfect separation with a little tailing of each zone (Fig. 2). Our expectations of the technique were at last realized and by treating the Kieselguhr support with alcoholic alkali, we suppressed the tailing.

After much thought, the penny dropped when A. J. P. remembered that short chain fatty acids were associated as dimers in concentrated solution. We then realized that in the zone moving down the column the low concentration edge would have its partial pressure proportional to the solute concentration, but in high concentrations the partial pressure would be proportional to the square root of the solute concentration. This would continuously sharpen the back of the zone and lengthen the front of the zone. Experimental proof was soon forthcoming with the incorporation of stearic acid in the stationary place when a volatile acid – stearic acid complex would form at all concentrations of the soluble acid and hence its partial pressure would be proportional to its concentration – this produced the originally expected symmetrical zones.

The way forward was now clear to open up the potential of the new method. We replaced the dropping burette with a novel screw burette and then with the first automatically recording burette. The column was heated by a simple vapour jacket, since electric heating to sufficient accuracy